BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LARKANA

INTERMEDIATE (11TH CLASS) EXAMINATION, 2017

CHEMISTRY

Past Paper 2017

Retyped faithfully from the original board paper — same questions, no changes.

Part –A: Multiple Choice Questions (MCQs Type A)

Time Allowed: 20 Minutes | **Maximum Marks:** 16

Note: Instruction: Identify the correct answer of each (MCQ) from its given four options and write it on the answer script/copy. Dated: 03 May 2017. Objective Types B and C contain the same 16 questions in a different order.

Q1. The precipitation occurs if the ionic concentration is.....

(A) Less than KSP
(B) Equal to KSP
(C) More than KSP
(D) None

Q2. CH3 COOH is a weak acid what is not true about it.

(A) Its dissociation in water is 50%
(B) It forms salts with bases
(C) Both
(D) None

Q3. Which of the following molecules has a net dipole moment.

(A) CO2
(B) CS2
(C) Both
(D) None

Q4. Sp3 orbitals are

(A) Coplanar
(B) Non planar
(C) Linear
(D) None

Q5. CaCO3 occurs in.....shape.

(A) Orthorhombic
(B) Trigonal
(C) Both
(D) None

Q6. For given process, heat changes at constant pressure and volume are related to each other.

(A) ΔE=q+w
(B) E=q-w
(C) Δ H=ΔE
(D) None

Q7. Pressure volume work is.....

(A) P Δ V
(B) fxd
(C) H+P Δ v
(D) None

Q8. Spectrum given due to transition of an electron from M to L shell is.....

(A) Absorption
(B) Emission
(C) Both
(D) None

Q9. Which of the following contains six bonding electrons.....

(A) C2H2
(B) NCl3
(C) Both
(D) None

Q10. Which of the following molecules has unpaired electrons in anti bonding molecular orbitals.

(A) O2
(B) N2
(C) Both
(D) None

Q11. If Kc is greater than unity then reaction is going in the direction...

(A) Forward
(B) Backward
(C) Equilibrium
(D) None

Q12. Chadwick was awarded Noble prize in.....

(A) Physics
(B) Chemistry
(C) Biology
(D) None

Q13. .....predicted in 1920 that some kind of neutral particles having mass equal to proton must be present in an atom.

(A) Chadwick
(B) Rutherford
(C) Dalton
(D) None

Q14. Cathode rays can produce X-rays when they strike an anode particularly with .....atomic mass.

(A) Large
(B) Small
(C) Both
(D) None

Q15. The molecular volume of CO2 is maximum at.....

(A) 127°C and 1 atm
(B) 0°C and 1 atm
(C) 273°C and 1 atm
(D) None

Q16. The deviation of a gas from ideal behavior is maximum at.....

(A) -10°C and 2.0 atm

(B) 100°C and 2.0 atm

(C) -10°C and 5.0 atm

(D) None

Part –B:- Constructed Response Questions(CRQs)/ (Short Answer Questions) (45 Marks)

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 69

> Note: Attempt any NINE of the following questions. Dated: 03 May 2017. Maximum marks 69 is the total for Part B and Part C together as printed on the subjective sheet.

Q1. Polar liquids have higher boiling point than comparable non-polar liquids. Explain.

Q2. What will be volume of N2 gas of 1.4 gram at S.T.P?

Q3. Define Graham's Law of diffusion? Why H2 has faster rate of diffusion than Methane gas?

Q4. Why are cations attracted towards cathode?

Q5. What is difference between Evaporation and vaporization?

Q6. Find out energy of electron in 2nd orbit of Hydrogen gas.

Q7. Why H F is weaker acid than HCL?

Q8. What is difference between Kc and Kp? Where both are equal or not?

Q9. What will be Molarity of NaOH weighed mass 30 g in 750 Cm3 solution?

Q10. Define the pH and What will be concentration of HCL having pH=1.3?

Q11. Why Ionic crystals are highly brittle?

Part –C:- (Extended Response Questions(ERQs)/(Long Answer Questions) (24 Marks)

> Note: Attempt any TWO of the following questions.

Q12.

(a) What is molecular formula? How it is determined?

(b) An organic compound contains C, H and O was found to have 75.2% of carbon, 10.15% Hydrogen. Its molecular mass is 115 gram. Calculate the molecular formula.

Q13.

(a) What is solubility product? Give its applications.

(b) Calculate the number of molecules in:-

(i) 1 Litre of H_2 at S.T.P.

(ii) 2.0 Litres of N_2 at S.T.P.

Q14.

(a) What is oxidation number? Describe the rules which are used to determine the oxidation number.

(b) What will be volume of hydrogen librated at standard temperature and pressure (STP) when sulphuric acid chemically acts on 120 grams of metallic magnesium Equation $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$?